

Alternative Data and Applications

MFE5340: AI in FE: Quantitative Investment

Chuan Shi
School of Data Science

Spring 2026



香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen



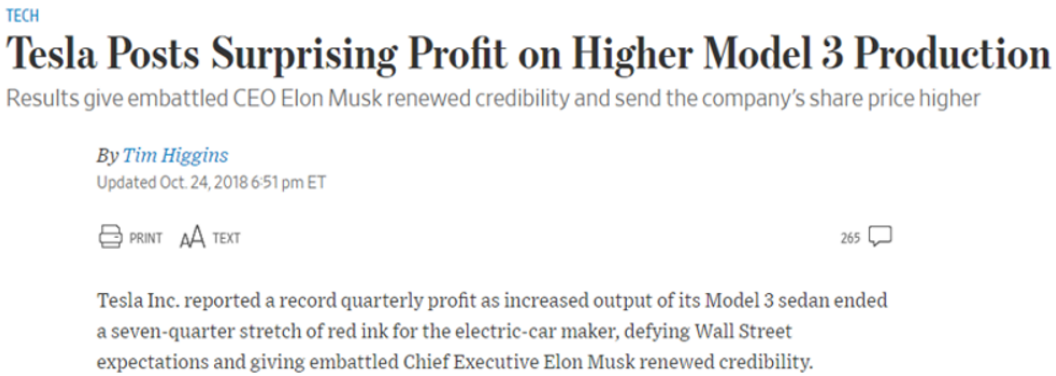
Table of Contents

1 Introduction

- Introduction
- Alternative Data and Quantitative Investment
- (Pseudo-)Alternative Data – Economic Links
- (Pseudo-)Alternative Data – Fund-Implied Information
- Implication for Financial Forecast
- Use Case: Patent Data in China

Exampe: Tesla's "Eye in the Sky"

- On October 25, 2018, Tesla's stock price rose by 9.14% following the release of its Q3 2018 earnings report, which significantly exceeded Wall Street expectations.
- The production of Tesla's Model 3 nearly doubled over the previous quarter.



TECH

Tesla Posts Surprising Profit on Higher Model 3 Production

Results give embattled CEO Elon Musk renewed credibility and send the company's share price higher

By Tim Higgins
Updated Oct. 24, 2018 6:51 pm ET

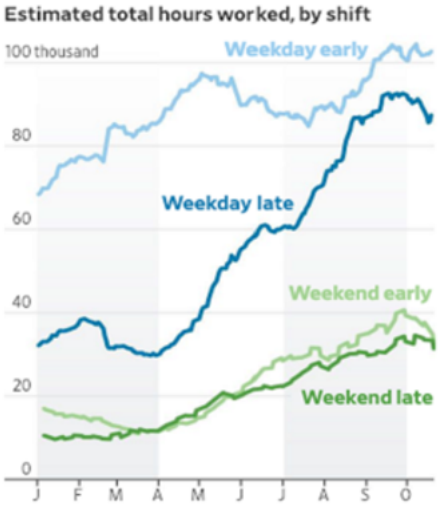
PRINT AA TEXT 265

Tesla Inc. reported a record quarterly profit as increased output of its Model 3 sedan ended a seven-quarter stretch of red ink for the electric-car maker, defying Wall Street expectations and giving embattled Chief Executive Elon Musk renewed credibility.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Exampe: Tesla's "Eye in the Sky"

- Before the earnings report, Thasos, an alternative company, and its clidents leveraged alternative data to predict Tesla's increased production, positioning themselves in the market ahead of the news.
- Thasos circled Tesla's 370 acres factory in Fremont, Calif., on an online map, creating a digital corral to isolate smartphone location signals that emanated from within it.
- It leased databases of trillions of geographic coordinates collected by smartphone apps, set its computers to find the pings created at Tesla's factory, showing the overnight shift swelled 30% from June to October.



© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Exampe: Tesla's "Eye in the Sky"

- Thasos leveraged unconventional data (smartphone signals) to gauge factory activity.
- Hedge fund clients acted on this insight to anticipate Tesla's Q3 performance.

Alternative data plays a crucial role in improving predictive capabilities for quantitative investing.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



What Is Alternative Data

- So, what is alternative data?
- Well, the “textbook” definition is:

> **Alternative data is a term for non-traditional data sets that are used to analyze financial markets and companies.**

- Let’s make it concrete.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



- Before **CRSP**, Wall Street was rife with tales of overnight fortunes. CRSP's advent shifted focus to quantitative research, making Eugene Fama's efficient market hypothesis a paradigm.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



The Emergence and Adoption of New Data

- Before **CRSP**, Wall Street was rife with tales of overnight fortunes. CRSP's advent shifted focus to quantitative research, making Eugene Fama's efficient market hypothesis a paradigm.

- Prior to **Compustat**, skepticism surrounded the link between fundamentals and stock prices. Compustat data enabled Ball and Brown to reveal the strong connection between earnings and stock prices through event studies, elevating accounting from bookkeeping to analyzing financial information's impact on capital markets.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Before **CRSP**, Wall Street was rife with tales of overnight fortunes. CRSP's advent shifted focus to quantitative research, making Eugene Fama's efficient market hypothesis a paradigm.

- Prior to **Compustat**, skepticism surrounded the link between fundamentals and stock prices. Compustat data enabled Ball and Brown to reveal the strong connection between earnings and stock prices through event studies, elevating accounting from bookkeeping to analyzing financial information's impact on capital markets.

- Before the **Odean dataset** (retail investor data), anomalies were attributed to risk compensation. Behavioral finance, leveraging Odean data, offered fresh insights into anomaly causes.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Before **CRSP**, Wall Street was rife with tales of overnight fortunes. CRSP's advent shifted focus to quantitative research, making Eugene Fama's efficient market hypothesis a paradigm.
- Prior to **Compustat**, skepticism surrounded the link between fundamentals and stock prices. Compustat data enabled Ball and Brown to reveal the strong connection between earnings and stock prices through event studies, elevating accounting from bookkeeping to analyzing financial information's impact on capital markets.
- Before the **Odean dataset** (retail investor data), anomalies were attributed to risk compensation. Behavioral finance, leveraging Odean data, offered fresh insights into anomaly causes.
- Before **I/B/E/S**, research on stock covariates focused on fundamental and technical variables. I/B/E/S estimates became crucial for providing incremental information in asset pricing.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Each dataset—whether CRSP, Compustat, Odean, or I/B/E/S—was a novel resource for investors of its time. While well-known today, these were initially considered *alternative* data.

- New data, along with new analytical methods like machine learning, continually reshapes and enhances our understanding of financial markets.

The inclusive acceptance and creative use of new data reflect a history of data-driven evolution in investment thought.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Table of Contents

2 Alternative Data and Quantitative Investment

- Introduction
- Alternative Data and Quantitative Investment
- (Pseudo-)Alternative Data – Economic Links
- (Pseudo-)Alternative Data – Fund-Implied Information
- Implication for Financial Forecast
- Use Case: Patent Data in China

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- **Academia:**
  - Alternative data opens new research avenues and methodologies.
  - Provides opportunities to publish in top-tier journals by exploring innovative topics.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- **Academia:**
  - Alternative data opens new research avenues and methodologies.
  - Provides opportunities to publish in top-tier journals by exploring innovative topics.

- **Industry:**
  - Traditional data signals are often already priced in, limiting alpha generation.
  - Alternative data offers unique insights and untapped sources of alpha.
  - Aims to gain a competitive edge through novel data-driven signals and predictive models.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Data Type	Representative Papers
Trading account	Barber and Odean (2000), Boehmer et al. (2008), Kaniel et al. (2008), Kelley and Tetlock (2013), Barrot et al. (2016)
Order book	Goettler et al. (2009), Cont et al. (2014)
Social media	Antweiler and Frank (2004), Tetlock (2007), Chen et al. (2014), Jame et al. (2016), Bartov et al. (2018), Leung et al. (2019), Umar (2022)
Crowdsourced data	Green et al. (2019), Da et al. (2021)
Text data	Cohen et al. (2020), Bekkerman et al. (2023), Bybee et al. (2023), Garcia et al. (2023)
Image data	Zhu (2019), Obaid and Pukthuanthong (2022), Jiang et al. (2023)
Weather data	Hirshleifer and Shumway (2003), Goetzmann et al. (2014)
Web search data	Da et al. (2011), Drake et al. (2012), Wu and Brynjolfsson (2015)

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- **Study:** Green et al. (2019) used Glassdoor data to analyze the link between employee ratings and stock returns.
- **Method:**
  - Categorized stocks into three groups (top 20%, middle 60%, bottom 20%) based on changes in employee ratings.
  - Constructed a high-minus-low factor to capture the effect of rating changes.
- **Findings:**
  - Significant excess returns for both equal-weighted and value-weighted portfolios.
  - Fama-MacBeth regression confirmed the findings.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Panel A: Future portfolio returns sorted by ΔRating

	Equal-weighted portfolios		Value-weighted portfolios	
	Average return	4-Factor alpha	Average return	4-Factor alpha
Low ΔRating	0.83	−0.24	0.59	−0.38**
	(1.28)	(−1.15)	(1.03)	(−2.02)
Middle group	1.06*	0.01	0.89*	0.02
	(1.77)	(0.07)	(1.83)	(0.21)
High ΔRating	1.66**	0.65***	1.33***	0.40*
	(2.50)	(2.62)	(2.62)	(1.81)
High – Low	0.84***	0.88***	0.74***	0.77***
	(2.67)	(2.70)	(3.03)	(3.26)

Panel B: Average portfolio characteristics

	ΔRating	β^{MKT}	Size (log)	BM	MOM
Low ΔRating	−0.50	1.24	22.91	0.63	7.63
Middle group	0.00	1.21	23.52	0.66	11.08
High ΔRating	0.53	1.25	22.98	0.62	10.28

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Challenge 1 – Unbiasedness

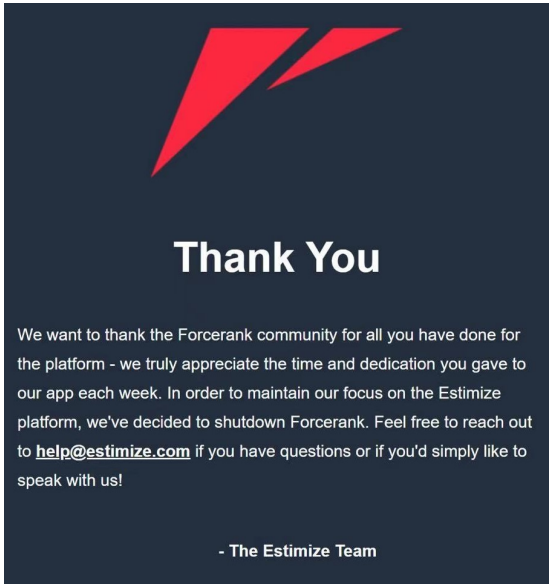
- Reliability Concerns with Glassdoor Data:
  - Lack of employee verification—anyone can leave a rating.
  - Negative bias—employees more likely to rate when dissatisfied.
  - Exaggeration—many 1-star or 5-star reviews.
  - Unclear rating system—subjective star allocation without clear standards.
  - Incentivized reviews—some companies reward employees for positive ratings.
- Possible over-representation of tech companies and under-representation of traditional sectors.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Da, Huang, and Jin (2021) used weekly stock ratings from retail investors on the Forcerank App to study the negative correlation between investors' extrapolative beliefs and future stock returns.
- Due to data limitations, the study sample only spans from February 2016 to December 2017, covering fewer than 300 stocks and about 1,000 users.
- The authors admitted that these users are not representative of all investors trading these stocks.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Da, Huang, and Jin (2021) used weekly stock ratings from retail investors on the Forcerank App to study the negative correlation between investors' extrapolative beliefs and future stock returns.
- Due to data limitations, the study sample only spans from February 2016 to December 2017, covering fewer than 300 stocks and about 1,000 users.
- The authors admitted that these users are not representative of all investors trading these stocks.



© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Challenge 2 – Missing Data

	S1	S2	S3	···	S2998	S2999	S3000
T1							
T2							
T3							
···							
T8							
T9							
T10							



	S1	S2	S3	···	S2998	S2999	S3000
T1							
T2							
T3							
···							
T8							
T9							
T10							



	S1	S2	S3	···	S2998	S2999	S3000
T1							
T2							
T3							
···							
T8							
T9							
T10							

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Much of alternative data is unstructured data.
- The use of machine learning has made textual data analysis a major area of research in academia.
- Cohen et al. (2020) analyzed whether changes in the wording of quarterly and annual reports of U.S. public companies are related to stock returns.
- Portfolio sort test confirms the following:

Companies with fewer wording changes tend to have higher future returns.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Panel A: Equally Weighted

	Sim_Cosine						*Sim_Jaccard*					
	Q1	Q2	Q3	Q4	Q5	Q5 – Q1	Q1	Q2	Q3	Q4	Q5	Q5 – Q1
Excess return	0.63*	0.72*	0.72**	0.85**	0.92***	0.31***	0.59	0.67*	0.69*	0.82**	0.98***	0.38***
	(1.68)	(1.96)	(2.11)	(2.59)	(2.80)	(3.13)	(1.48)	(1.74)	(1.89)	(2.35)	(3.01)	(2.65)
Three-factor alpha	−0.15**	−0.08	−0.05	0.09	0.18***	0.34***	−0.16**	−0.10	−0.06	0.08	0.28***	0.44***
	(−2.19)	(−1.10)	(−0.72)	(1.21)	(2.66)	(4.45)	(−1.99)	(−1.22)	(−0.81)	(1.05)	(3.47)	(4.56)
Five-factor alpha	−0.12*	−0.05	−0.04	0.10	0.21***	0.32***	−0.14*	−0.07	−0.06	0.09	0.28***	0.42***
	(−1.75)	(−0.74)	(−0.53)	(1.29)	(3.28)	(4.21)	(−1.84)	(−0.93)	(−0.86)	(1.19)	(3.57)	(4.31)
	Sim_MinEdit						*Sim_Simple*					
	Q1	Q2	Q3	Q4	Q5	Q5 – Q1	Q1	Q2	Q3	Q4	Q5	Q5 – Q1
Excess return	0.61	0.66*	0.70*	0.86**	0.99***	0.36***	0.72*	0.79**	0.82**	0.90***	0.90***	0.18
	(1.60)	(1.78)	(1.94)	(2.58)	(3.36)	(2.69)	(1.87)	(2.12)	(2.34)	(2.73)	(3.04)	(1.20)
Three-factor alpha	−0.19**	−0.14*	−0.10	0.10	0.30***	0.48***	−0.08	−0.02	0.03	0.14**	0.20**	0.28***
	(−2.56)	(−1.91)	(−1.52)	(1.37)	(4.00)	(5.96)	(−1.09)	(−0.21)	(0.38)	(2.01)	(2.57)	(3.22)
Five-factor alpha	−0.15**	−0.11	−0.08	0.12*	0.30***	0.45***	−0.06	0.03	0.04	0.16**	0.21***	0.27***
	(−2.14)	(−1.59)	(−1.31)	(1.70)	(4.11)	(5.46)	(−0.89)	(0.37)	(0.63)	(2.30)	(2.68)	(3.01)

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Common causes of wording changes:
  - More negative sentiment
  - Higher uncertainty
  - Increased litigiousness
  - CEO/CFO turnover

- These factors often indicate higher risks and uncertainties in operations.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Challenge 4 – Domain Knowledge

- Alternative data often requires deep domain knowledge to extract meaningful insights from it.
- Domain knowledge enables a deep understanding of the data's origin, the underlying processes, and its financial relevance.
- Customized, raw data allows users to perform their own cleaning, analysis, and back-testing, leading to unique and potentially less crowded investment signals.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Challenge 4 – Domain Knowledge

- Alternative data often requires deep domain knowledge to extract meaningful insights from it.
- Domain knowledge enables a deep understanding of the data's origin, the underlying processes, and its financial relevance.
- Customized, raw data allows users to perform their own cleaning, analysis, and back-testing, leading to unique and potentially less crowded investment signals.

> Large asset management firms: "What we're generally looking for is something a little bit more raw, a bit more unprocessed, where we can really understand what the data is."

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Table of Contents

3 (Pseudo-)Alternative Data – Economic Links

- Introduction
- Alternative Data and Quantitative Investment
- (Pseudo-)Alternative Data – Economic Links
- (Pseudo-)Alternative Data – Fund-Implied Information
- Implication for Financial Forecast
- Use Case: Patent Data in China

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Informal definition for subsequent discussion:
  - Not typical firm characteristics or macroeconomic data
  - Easier to obtain than alternative data
  - Often overlooked and underutilized

- Examples:
  - Economic links between companies, or inter-firm links (e.g., the T-link you have seen in the midterm!)
  - Implied information from mutual funds (fund performance, fund flow, etc)

- Credits: I thank my long-term collaborator, Professor Liu Yangyi (Southwest Jiaotong University), for the discussion and preparation of the materials in this section.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Inter-firm links refer to direct or indirect relationships between firms, often based on factors such as:
  - Supply chains
  - Industry relations
  - Partnerships
  - Shared resources

Source	Inter-Firm Links
Hou (2007)	intra-industry links
Cohen and Frazzini (2008)	key customer links
Cohen and Lou (2012)	complicated firm links
Lee et al. (2019)	technological links
Parsons et al. (2020)	geographic links
Ali and Hirshleifer (2020)	shared analyst coverage links

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Inter-Firm Links and Lead-Lag Effect

- Economically linked firms can influence each other's performance and reflect similar market dynamics. This is often seen in stock returns through a **lead-lag effect**, where one group of assets responds to information or events faster than another.

- The lead-lag effect represents the uneven diffusion of information in the market, where leading firms' returns can serve as predictors for lagging firms.

- Terminology:
  - **Focal Firms:** Companies whose returns are being predicted.
  - **Linked Firms:** Economically linked firms that act as predictors.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- The first step in empirical research on the lead-lag effect is to measure inter-firm links from an economic perspective. Once such a link is established, we can construct a factor (variable) as follows:

$$x_{i,t} = \frac{\sum_{j \in \mathcal{S}} \text{LINK}_{i,j,t} \times \text{RET}_{j,t}}{\sum_{j \in \mathcal{S}} \text{LINK}_{i,j,t}},$$

  where
  - $\mathcal{S}$: Set of linked firms to focal firm i at time t.
  - $\text{LINK}_{i,j,t}$: Degree of connection between focal firm i and linked firm j.
  - $\text{RET}_{j,t}$: Return of linked firm j at time t.

- Portfolios constructed based on inter-firm links yield significant excess returns.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- The higher the technology-linked returns, the higher the return of the focal company in the following month.

Panel A: Portfolio returns

Decile	Excess returns (%)	CAPM alpha (%)	3-Factor alpha (%)	4-Factor alpha (%)	5-Factor alpha (%)	6-Factor alpha (%)
1 (Low)	0.42 (1.46)	−0.16 (−1.04)	−0.33 (−2.66)	−0.13 (−1.09)	−0.27 (−2.17)	−0.12 (−0.99)
10 (High)	1.59 (5.38)	1.05 (5.37)	0.93 (6.41)	0.95 (6.36)	1.10 (8.11)	1.09 (7.97)
L/S (Equal-weights)	**1.17** (5.47)	**1.22** (5.70)	**1.26** (5.88)	**1.08** (4.98)	**1.37** (6.49)	**1.21** (5.76)
L/S (Value-weights)	**0.69** (3.19)	**0.74** (3.40)	**0.80** (3.62)	**0.65** (2.91)	**0.86** (3.81)	**0.73** (3.24)

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Cumulative returns rise within 6 Months:
  - Nature of technology-related innovation
  - Limited attention of investors
  - Limits to arbitrage
- Risk-based explanation difficult to establish
- Reference: Lee et al. (2019), JFE: *Technological Links and Predictable Returns*
- We will discuss this further as a use case with patent data in China.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Geographic links refer to the relationships between companies located in the same city but operating in different industries. These connections can impact stock relationships.
- References: Parsons et al. (2020), RFS, *Geographic Lead-Lag Effects*

Table 5
Area momentum trading strategy

Momentum trading strategy, quintiles

	Returns									
	Mean (%)	*CAPM α*	*t-stat*	*FF-3 α*	*t-stat*	*FF-3+MOM α*	*t-stat*	*FF-5 α*	*t-stat*	*Sharpe Ratio*
Lowest city return	0.735	−0.258	−3.228	−0.244	−3.011	−0.223	−2.713	−0.199	−2.333	0.207
	0.876	−0.077	−1.112	−0.038	−0.520	0.002	0.021	−0.016	−0.191	0.318
	1.027	0.109	1.432	0.127	1.583	0.178	2.309	0.132	1.553	0.452
	0.949	−0.009	−0.102	−0.002	−0.022	−0.024	−0.288	0.013	0.122	0.366
Highest city return	1.158	0.212	3.196	0.211	3.029	0.195	2.629	0.217	2.714	0.526
5-1 spread	0.423 [3.65]	0.471	[4.16]	0.455	[3.99]	0.419	[3.40]	0.417	[3.319]	

This table reports the performance of a trading strategy that exploits return continuation at the geographic level. Every month, we rank eac
firms headquartered in the same city, outside its industry. We then construct quintile value weighted portfolios of the sorted firms, and ho
every month. Displayed are mean returns, CAPM α, FF-3 α, FF-3 + Momentum α, and FF-5 α of each quintile portfolio. *Volatility* is the m
Mkt share is the proportional market share of the individual portfolios; *Size* is the natural logarithm of the market value of the portfolios (in
portfolios. Newey-West (5 lags) standard errors. Monthly data, 1971-2013. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Information about a company's key customers significantly impact its stock price, but investors often underreact to this information.

- Reference: Cohen and Frazzini (2008), JF, *Economic Links and Predictable Returns*

Panel A: Value Weights	Q1(Low)	Q2	Q3	Q4	Q5(High)	L/S
Excess returns	−0.596	−0.157	0.125	0.313	0.982*	1.578*
	[−1.42]	[−0.41]	[0.32]	[0.79]	[2.14]	[3.79]
Three-factor alpha	−1.062*	−0.796*	−0.541*	−0.227	0.493*	1.555*
	[−3.78]	[−3.61]	[−2.15]	[−0.87]	[1.98]	[3.60]
Four-factor alpha	−0.821*	−0.741*	−0.488	−0.193	0.556*	1.376*
	[−2.93]	[−3.28]	[−1.89]	[−0.72]	[1.99]	[3.13]
Five-factor alpha	−0.797*	−0.737*	−0.493	−0.019	0.440	1.237*
	[−2.87]	[−3.04]	[−1.94]	[−0.07]	[1.60]	[2.99]

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Each industry in which a diversified company operates can impact its stock returns, but investors may not respond promptly or accurately to this information.

- The predictor can be constructed as:

$$\text{COM RET}_{i,t} = \frac{\sum_{j \in \text{Ind}_i} w_{i,j,t} \text{RET}_{j,t}}{\sum_{j \in \text{Ind}_i} w_{i,j,t}},$$

  where:
  - $w_{i,j,t} = \frac{\text{Sales}_{i,j,t}}{\text{Sales}_{i,t}}$: the proportion of industry j's revenue in company i's total revenue.
  - $\text{RET}_{j,t}$: the return of the portfolio of companies in industry j at month t.

- References: Cohen and Lou (2012), JFE, *Complicated Firms*.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Decile	Excess returns	1-Factor alpha	3-Factor alpha	4-Factor alpha	5-Factor alpha
Panel A: Equal weights					
1	0.14%	−0.47%	−0.71%	−0.61%	−0.65%
(Low)	(0.43)	(−2.83)	(−4.80)	(−4.01)	(−4.39)
2	0.08%	−0.50%	−0.73%	−0.64%	−0.68%
	(0.28)	(−3.57)	(−5.94)	(−5.35)	(−5.90)
3	0.50%	−0.03%	−0.25%	−0.18%	−0.20%
	(1.85)	(−0.25)	(−2.30)	(−1.63)	(−1.85)
4	0.67%	0.14%	−0.09%	0.00%	−0.01%
	(2.48)	(1.11)	(−0.82)	(0.01)	(−0.09)
5	0.85%	0.34%	0.11%	0.18%	0.19%
	(3.26)	(2.83)	(1.16)	(1.90)	(1.96)
6	0.85%	0.32%	0.08%	0.15%	0.15%
	(3.20)	(2.72)	(0.84)	(1.54)	(1.50)
7	0.90%	0.37%	0.13%	0.15%	0.16%
	(3.38)	(3.11)	(1.36)	(1.43)	(1.57)
8	0.97%	0.44%	0.21%	0.22%	0.24%
	(3.63)	(3.67)	(2.15)	(2.00)	(2.20)
9	0.99%	0.46%	0.24%	0.24%	0.25%
	(3.66)	(3.61)	(2.23)	(2.12)	(2.12)
10	1.31%	0.74%	0.48%	0.47%	0.47%
(High)	(4.34)	(4.63)	(3.63)	(3.30)	(3.09)
L/S	**1.18%**	**1.21%**	**1.18%**	**1.08%**	**1.12%**
	(5.51)	(5.52)	(5.30)	(4.48)	(4.50)

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Use shared analyst coverage as a proxy to construct connection metrics.
- The predictor is defined as:

$$\text{CF RET}_{i,t} = \frac{\sum_j n_{i,j,t}\text{RET}_{j,t}}{\sum_j n_{i,j,t}}$$

  where $n_{i,j,t}$ is the number of analysts covering both firms i and j over the period from $t-11$ to t month.

- Reference: Ali and Hirshleifer (2020) , JFE, *Shared Analyst Coverage: Unifying Momentum Spillover Effects*

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Equal weighted

Quintile	Excess ret	4-factor alpha	5-factor alpha	Mkt-Rf	SMB	HML	UMD	ST reversal
1	−0.36	−1.05	−1.21	1.01	0.67	0.03	−0.05	0.75
	(−1.03)	(−6.46)	(−11.65)	(39.96)	(10.95)	(0.59)	(−1.51)	(17.21)
2	0.47	−0.23	−0.29	0.99	0.57	0.20	−0.04	0.33
	(1.66)	(−2.69)	(−4.12)	(48.54)	(10.54)	(5.59)	(−1.96)	(7.65)
3	0.82	0.15	0.14	0.96	0.53	0.29	−0.07	0.08
	(3.20)	(2.12)	(1.83)	(42.71)	(10.42)	(6.94)	(−2.98)	(1.74)
4	1.07	0.39	0.43	1.02	0.59	0.25	−0.07	−0.19
	(4.09)	(4.98)	(6.03)	(48.83)	(13.34)	(6.31)	(−2.92)	(−5.03)
5	1.41	0.76	0.89	1.08	0.85	0.11	−0.10	−0.65
	(4.66)	(5.11)	(8.93)	(43.02)	(14.94)	(2.60)	(−3.33)	(−14.97)
5–1	1.76	1.81	2.10	0.07	0.18	0.08	−0.06	−1.41
	(6.23)	(6.15)	(11.88)	(1.65)	(1.74)	(1.09)	(−1.02)	(−17.80)

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Shared Analyst Coverage – Even More

- Shared analysts coverage can explain other lead-lag effects.

	Mean return	4-factor alpha	5-factor alpha	4-factor + CF momentum factor alpha	5-factor + CF momentum factor alpha
	(1)	(2)	(3)	(4)	(5)
CF momentum factor	1.33	1.38	1.68		
	(4.70)	(4.65)	(9.67)		
Industry momentum factor	0.64	0.64	0.85	−0.24	−0.04
	(3.06)	(2.92)	(5.13)	(−2.06)	(−0.27)
Geographic momentum factor	0.31	0.31	0.43	−0.23	−0.23
	(2.23)	(2.47)	(3.91)	(−2.46)	(−2.34)
Customer momentum factor	1.15	0.97	1.29	0.43	0.74
	(3.44)	(2.47)	(3.62)	(1.09)	(1.79)
Supplier industry momentum factor	0.58	0.55	0.71	−0.26	−0.12
	(2.54)	(2.24)	(3.98)	(−1.73)	(−0.79)
Customer industry momentum factor	0.40	0.45	0.62	−0.42	−0.10
	(1.73)	(1.94)	(3.84)	(−2.86)	(−0.71)
Complicated firm momentum factor	0.42	0.40	0.54	−0.18	0.02
	(2.56)	(2.25)	(4.38)	(−1.41)	(0.17)
Technology momentum factor	0.55	0.50	0.72	−0.52	−0.60
	(2.06)	(1.89)	(3.24)	(−2.90)	(−2.82)

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

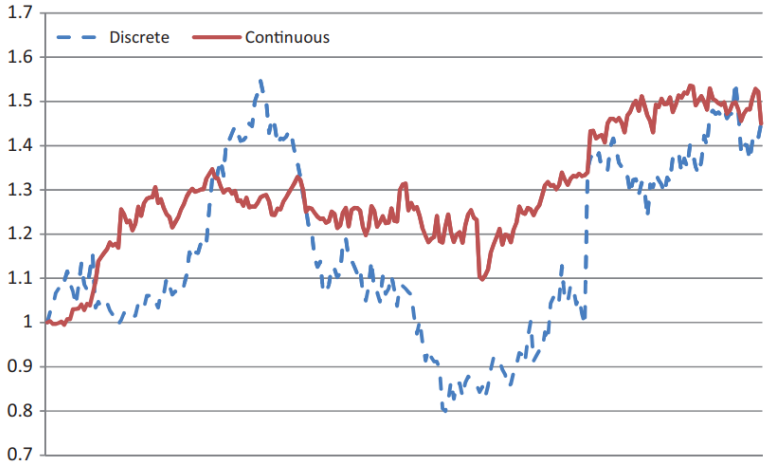
Shared Analyst Coverage – China Evidence

	Low	Middle	High	High − Low
Small	0.96 (1.30)	1.37 (1.86)	1.57 (2.07)	0.61 (2.68)
Big	0.46 (0.74)	0.70 (1.14)	1.28 (1.86)	0.82 (2.26)
Average	0.71 (1.08)	1.04 (1.59)	1.42 (2.05)	0.71 (2.68)

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Information Discreteness

- Investor decisions are influenced by the **representativeness of information**.
- **The more continuous the information, the weaker its representativeness,** resulting in less investor reaction and stronger momentum effects (Da et al. 2014, RFS).



© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Information Discreteness and Lead-Lag Effect

- Various lead-lag effects are also associated with the degree of information discreteness (Huang et al. 2022).
- When the information from leading firms is small and repeated (i.e., more continuous), investor limited attention issues are more pronounced, leading to higher excess returns from lead-lag effects.
- When the information from leading firms is highly visible (i.e., more discrete), the market fully absorbs the impact on lagging firms, and lead-lag excess returns are harder to achieve.

Information discreteness serves as a cognitive trigger that reduces investor inattention, irrespective of the nature of the economic link.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Taking customer link as an example, we can measure information discreteness of a linked firm as:

$$\mathrm{ID}_{j,t} = \mathrm{sign}(CR_{j,t}) \times (\%\mathrm{neg}_{j,t} - \%\mathrm{pos}_{j,t}),$$

where $CR_{j,t}$ is a customer firm j's cumulative return over the past three months, $\mathrm{sign}(CR_{j,t})$ is the sign of $CR_{j,t}$, and $\%\mathrm{neg}_{j,t}$ and $\%\mathrm{pos}_{j,t}$ are the percentage of days during the past three months with positive and negative returns, respectively.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Taking customer link as an example, we can measure information discreteness of a linked firm as:

$$\mathrm{ID}_{j,t} = \mathrm{sign}(CR_{j,t}) \times (\%\mathrm{neg}_{j,t} - \%\mathrm{pos}_{j,t}),$$

  where $CR_{j,t}$ is a customer firm j's cumulative return over the past three months, $\mathrm{sign}(CR_{j,t})$ is the sign of $CR_{j,t}$, and $\%\mathrm{neg}_{j,t}$ and $\%\mathrm{pos}_{j,t}$ are the percentage of days during the past three months with positive and negative returns, respectively.

- When $\%\mathrm{pos}_{j,t} \gg \%\mathrm{neg}_{j,t}$, ($\mathrm{sign}(CR_{j,t})$ is likely to be 1,) a past positive customer return is attributable to many small positive returns. According to the formula, the ID will be small.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Taking customer link as an example, we can measure information discreteness of a linked firm as:

$$\mathrm{ID}_{j,t} = \mathrm{sign}(CR_{j,t}) \times (\%\mathrm{neg}_{j,t} - \%\mathrm{pos}_{j,t}),$$

  where $CR_{j,t}$ is a customer firm j's cumulative return over the past three months, $\mathrm{sign}(CR_{j,t})$ is the sign of $CR_{j,t}$, and $\%\mathrm{neg}_{j,t}$ and $\%\mathrm{pos}_{j,t}$ are the percentage of days during the past three months with positive and negative returns, respectively.

- When $\%\mathrm{pos}_{j,t} \gg \%\mathrm{neg}_{j,t}$, ($\mathrm{sign}(CR_{j,t})$ is likely to be 1,) a past positive customer return is attributable to many small positive returns. According to the formula, the ID will be small.

- When $\%\mathrm{neg}_{j,t} \gg \%\mathrm{pos}_{j,t}$, ($\mathrm{sign}(CR_{j,t})$ is likely to be -1,) a past negative customer return is attributable to many small negative returns. According to the formula, the ID will also be small.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

	CR1 (Low)	CR2	CR3 (High)	CR 3 - 1
Panel A: Avg Monthly Excess Returns				
ID1 (Discrete)	0.30	0.49	0.50	0.21
	(0.84)	(1.67)	(1.48)	(1.10)
ID2	0.22	0.54	0.83***	0.61***
	(0.61)	(1.84)	(2.61)	(3.17)
ID3 (Continuous)	0.17	0.33	1.15***	0.98***
	(0.49)	(1.00)	(3.77)	(4.65)
ID 3 - 1	−0.13	−0.16	0.65***	0.78***
	(−0.76)	(−1.26)	(3.54)	(3.24)
Panel B: Avg Monthly FF5 Alpha				
	CR1 (Low)	CR2	CR3 (High)	CR 3 - 1
ID1 (Discrete)	−0.24	−0.03	−0.21	0.03
	(−1.54)	(−0.32)	(−1.56)	(0.12)
ID2	−0.37***	−0.04	0.20	0.57***
	(−2.69)	(−0.37)	(1.74)	(2.95)
ID3 (Continuous)	−0.37***	−0.21	0.56***	0.93***
	(−2.62)	(−1.35)	(3.20)	(4.29)

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Economic Links – Summary

- Investors lack sufficient attention to inter-firm link information, leading to underreaction.
- Various inter-firm links provide significant predictive power for future changes in company fundamentals.
- Most inter-firm link data is publicly available.
- Empirical evidence suggests that lead-lag effects due to inter-firm links are widely effective in the Chinese stock market.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Table of Contents

4 (Pseudo-)Alternative Data – Fund-Implied Information

- Introduction
- Alternative Data and Quantitative Investment
- (Pseudo-)Alternative Data – Economic Links
- (Pseudo-)Alternative Data – Fund-Implied Information
- Implication for Financial Forecast
- Use Case: Patent Data in China

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Measure fund skill as the weighted average of its stock holdings' alphas:

$$S_{i,t+1} = \sum_{n=1}^{N_t^s} w_{n,t}^i \alpha_{n,t+1},$$

where:

- $S_{i,t+1}$: Skill of fund i at time $t+1$
- $w_{n,t}^i$: Weight of stock n in fund i's portfolio at time t
- $\alpha_{n,t+1}$: Alpha of stock n at time $t+1$

- The goal is to predict $\alpha_{n,t+1}$ based on information available at time t.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Assume that we can form a noisy estimate of fund skill $S_{i,t+1}$ using data up to period t:

$$S_{i,t+1} = \hat{S}_{i,t} + e_{i,t+1}.$$

- In practice, we can use fund's abnormal return against some factor model over some a number of T periods in the past to compute $\hat{S}_{i,t}$.

- With $\hat{S}_{i,t}$, in matrix form, we have:

$$\hat{\boldsymbol{S}} = \mathbf{W}\boldsymbol{\alpha} + \boldsymbol{e},$$

where:

  - $\hat{\boldsymbol{S}}$: Vector of estimated fund skills
  - $\mathbf{W}$: $N_t \times N_t^s$ matrix of fund holding weights (N_t: number of funds; N_t^s: number of stocks)
  - $\boldsymbol{\alpha}$: Vector of stock alphas
  - $\boldsymbol{e}$: Vector of error terms

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- In theory, stock alphas can be estimated with OLS:

$$\hat{\boldsymbol{\alpha}}_{\text{OLS}} = \left(\mathbf{W}^\top \mathbf{W}\right)^{-1} \mathbf{W}^\top \hat{\boldsymbol{S}}.$$

- However, there are more stocks than funds, making OLS inapplicable.

- **Solution:** Use generalized inverse:

$$\hat{\boldsymbol{\alpha}}_{\text{GI}} = \left(\mathbf{V}\hat{\mathbf{D}}\mathbf{V}^\top\right)^{-1} \mathbf{W}^\top \hat{\boldsymbol{S}},$$

where:

— $\mathbf{V}$: Eigenvector matrix of $\mathbf{W}^\top \mathbf{W}$

— $\hat{\mathbf{D}}$: Modified diagonal matrix based on eigenvalues

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Fund-Implied Stock Alpha – Empirical Results

Performance of GIA estimated using fund holdings

	Characteristic Quintile Ranks			Net Return (%)				
	SIZE	BM	MOM	Q1	Q2	Q3	Q4	JT4
D1 (Bottom)	3.53	2.81	2.97	3.02	3.02	2.95	3.03	2.97
D2	2.85	3.01	2.89	3.31	3.33	3.22	3.27	3.25
D3	2.41	3.06	2.85	3.24	3.52	3.37	3.18	3.32
D4	1.98	3.14	2.87	3.19	3.21	3.27	3.45	3.25
D5	1.56	3.21	2.93	3.64	3.82	3.68	3.73	3.70
D6	1.69	3.2	2.96	3.69	3.59	3.53	3.56	3.59
D7	2.07	3.14	2.94	3.70	3.68	3.39	3.64	3.62
D8	2.44	3.08	2.97	3.88	3.73	3.75	3.58	3.74
D9	2.89	2.97	3.07	4.17	3.92	3.62	3.49	3.81
D10 (Top)	3.51	2.7	3.26	4.55	4.34	3.93	3.42	4.07
D10–D1				1.53	1.31	0.98	0.39	1.10
				(5.11)	(4.08)	(2.85)	(0.65)	(3.99)

- References: Wermers et al. (2012), RFS, *Forecasting Stock Returns through an Efficient Aggregation of Mutual Fund Holdings*

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Fund managers should buy or sell stocks in proportion to portfolio weights based on net inflows or outflows.
- The flow-induced trading (FIT) indicator for each stock can be calculated using fund flows and fund holdings for each t.
- We can predict FIT for $t+1$ for each stock, i.e., $\mathbb{E}[FIT_n]$, by using historical data.
- $\mathbb{E}[FIT_n]$ predicts future stock returns up to 4 quarters.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

The expected flow-induced price effect

Panel A: Stocks ranked by $E[FIT]$

Decile	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	Excess Return	3-Factor Alpha	Excess Return	3-Factor Alpha
	Qtr. 1			Qtr. 1–4			Qtr. 5		Qtr. 6–8		Qtr. 6–12	
1	0.38%	−0.50%	−0.25%	0.53%	−0.40%	−0.13%	0.52%	−0.24%	0.84%	−0.01%	0.94%	0.13%
10	1.21%	0.43%	0.27%	0.97%	0.18%	0.24%	0.63%	−0.01%	0.54%	−0.23%	0.67%	−0.14%
10 - 1	**0.84%**	**0.93%**	**0.53%**	**0.44%**	**0.58%**	**0.37%**	0.10%	0.23%	**−0.29%**	−0.22%	**−0.27%**	**−0.27%**
	(3.96)	(4.15)	(3.51)	(2.63)	(3.30)	(2.26)	(0.49)	(0.81)	(−2.06)	(−1.32)	(−2.17)	(−2.04)

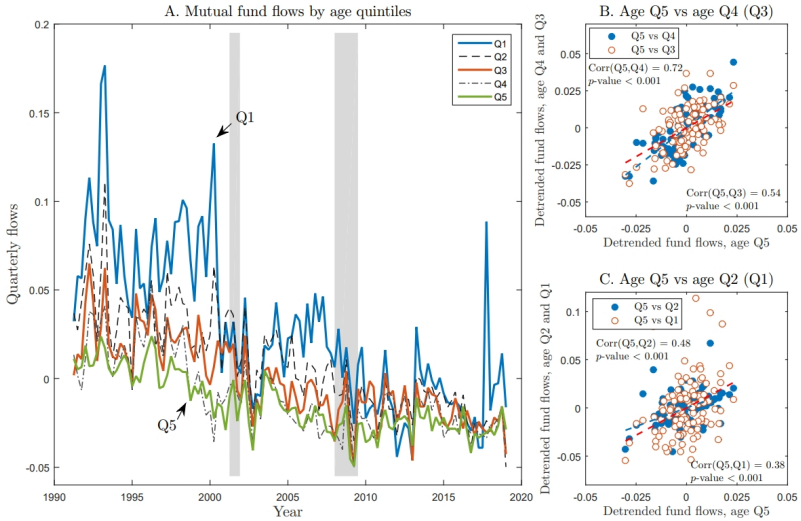
Panel B: Mutual funds ranked by $E[FIT^*]$

Decile	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	4-Factor Alpha	Excess Return	3-Factor Alpha	Excess Return	3-Factor Alpha	Excess Return	3-Factor Alpha
	Qtr. 1			Qtr. 1–4			Qtr. 5		Qtr. 6–8		Qtr. 6–12	
1	0.58%	−0.17%	−0.03%	0.62%	−0.15%	−0.11%	0.71%	0.07%	0.87%	0.25%	0.79%	0.18%
10	1.14%	0.54%	0.37%	1.02%	0.40%	0.26%	0.70%	0.11%	0.56%	−0.05%	0.57%	−0.07%
10 - 1	**0.55%**	**0.71%**	**0.41%**	**0.40%**	**0.55%**	**0.37%**	−0.01%	0.04%	**−0.31%**	**−0.30%**	−0.22%	−0.25%
	(2.66)	(3.14)	(2.58)	(2.65)	(3.44)	(2.34)	(−0.06)	(0.22)	(−2.23)	(−2.07)	(−1.70)	(−1.87)

- References: Lou (2012), RFS, *A Flow-Based Explanation for Return Predictability*

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

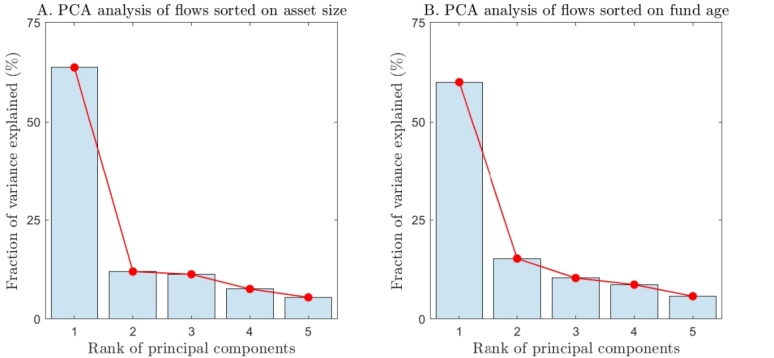
- Influenced by macroeconomic uncertainty, fund flows exhibit a significant common component.



© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Common Fund Flow

- There is one dominant factor that drives much of the common variation of fund flow shocks — the first principal component.
- It explains over 60% of the variance.



Note: Panel A plots the fraction of variance explained by different principal components from the principal component analysis (PCA) of flows sorted on asset size. Panel B plots the fraction of variance explained by different principal components from the PCA of flows sorted on fund age presented in panel A.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Using past 36 months of data and a time series regression to estimate common fund flow beta:

$$R_{i,t} = a_{i,t} + \beta_{i,t}^{flow} \times \text{common_flow}_t + e_{i,t}.$$

- Fund managers tend to hold stocks with **low flow beta** to hedge its exposure to common fund flow.
- Demand for stocks with low common fund flow beta leads to higher prices and lower future returns. The reverse is true for high common fund flow beta stocks.
- Common fund flow beta has a **positive** risk premium.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Table 3: Excess returns and CAPM alphas of portfolios sorted on flow betas.

	Panel A. CRSP mutual funds alone		Panel B. CRSP-Morningstar intersection	
β_i^{flow} quintiles	Excess returns	CAPM α	Excess returns	CAPM α
Q1	5.22	−5.51**	4.84	−4.90**
	[1.22]	[−2.56]	[1.26]	[−2.58]
Q2	7.17**	−0.97	7.75***	0.51
	[2.34]	[−0.79]	[2.87]	[0.49]
Q3	8.18***	0.57	7.93***	0.03
	[2.88]	[0.52]	[2.77]	[0.03]
Q4	9.27***	1.05	10.72***	1.79
	[3.13]	[1.20]	[3.24]	[1.44]
Q5	12.02***	2.10	13.34***	1.80
	[3.02]	[1.02]	[2.83]	[0.70]
Q5 − Q1	6.81**	7.62**	8.50**	6.70**
	[2.19]	[2.42]	[2.56]	[2.02]

- References: Dou et al. (2020), WP, *Common Fund Flows: Flow Hedging and Factor Pricing*

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Table of Contents

5 Implication for Financial Forecast

- Introduction
- Alternative Data and Quantitative Investment
- (Pseudo-)Alternative Data – Economic Links
- (Pseudo-)Alternative Data – Fund-Implied Information
- Implication for Financial Forecast
- Use Case: Patent Data in China

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Question: Does alternative data enhance forecast accuracy in finance?

- In academic research on expected stock returns, the answer is often yes.
- However, how does alternative data affect the accuracy of sell-side analysts' earnings forecasts?
- Dessaint et al. (2022) explored this unique angle, providing both theoretical insights and empirical evidence.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Short-Term Oriented Nature of Alternative Data

- Analysis of 26 academic papers shows that alternative data sources (e.g., social media, satellite data) generally have a forecast horizon of less than one year.
- Alternative data is thus short-term oriented, offering information incremental primarily for short-term forecasts.
- Key Questions:
  - How does the influx of short-term data affect analysts' short-term vs. long-term earnings forecasts?
  - Can this improve overall forecast accuracy?

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Theoretical Model: Analyst Focus Shift

- Model assumes analysts allocate effort across short-term and long-term forecasts to minimize errors and costs.
- Alternative data lowers the cost of short-term forecasts and improves their accuracy.
- Result: Analysts shift more effort towards short-term predictions, potentially at the expense of long-term accuracy due to limited resources.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- For a given date t, analyst i, and time horizon h, gather all earnings forecasts made by that analyst (assume there are $j = 1, \ldots, J$ forecasts for different firms!). Then, identify the actual earnings from the financial reports and perform a **cross-sectional** regression of the actual earnings, standardized by total assets, on the standardized earnings forecasts:

$$\frac{\text{Earnings}_{jh}}{\text{Assets}_j} = \alpha + \beta \frac{\text{Forecasts}_{jh}}{\text{Assets}_j} + e_{jh}.$$

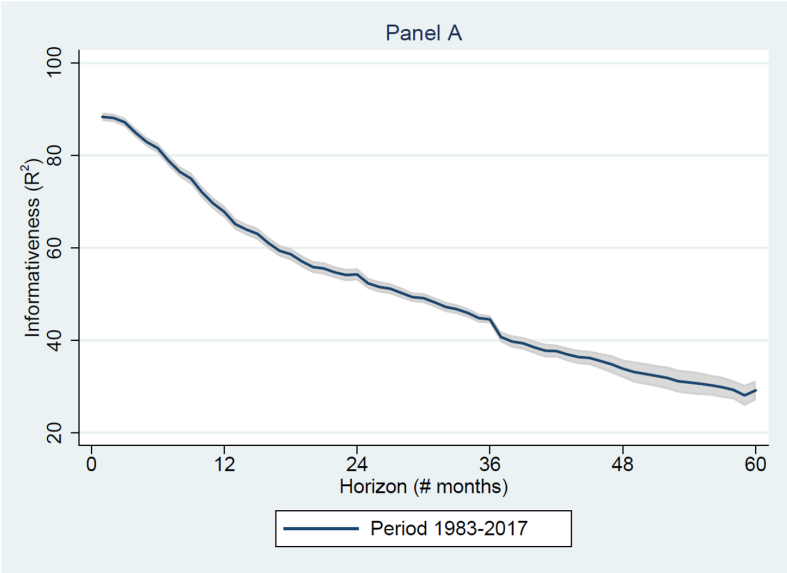
© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- For a given date t, analyst i, and time horizon h, gather all earnings forecasts made by that analyst (assume there are $j = 1, \ldots, J$ forecasts for different firms!). Then, identify the actual earnings from the financial reports and perform a **cross-sectional** regression of the actual earnings, standardized by total assets, on the standardized earnings forecasts:

$$\frac{\text{Earnings}_{jh}}{\text{Assets}_j} = \alpha + \beta \frac{\text{Forecasts}_{jh}}{\text{Assets}_j} + e_{jh}.$$

- The R-squared from the above regression is used as the measure of forecast accuracy for date t, analyst i, and time horizon h. Then, for a given time horizon h, aggregate the R-squared values across both time and analyst to obtain the average R-squared for h, representing the forecast accuracy level of analysts at time horizon h.

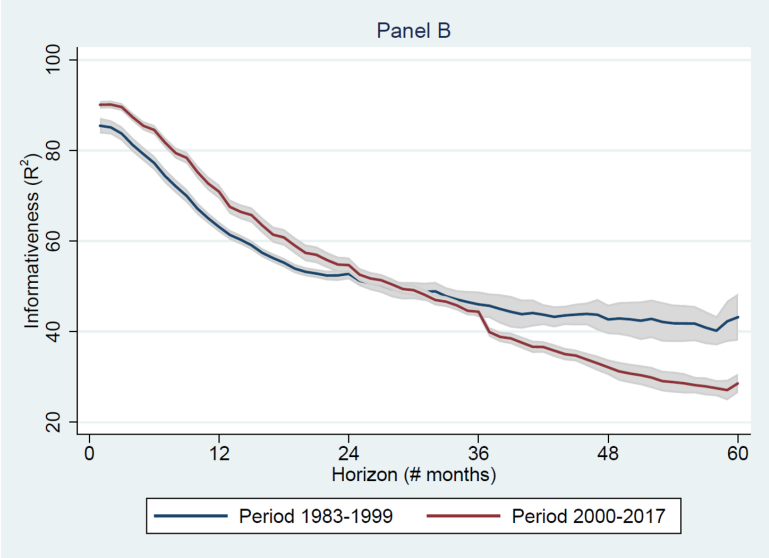
© 2025 – 2026 Chuan Shi | Alternative Data and Applications



出处: Dessaint, Foucault, and Fresard (2022)

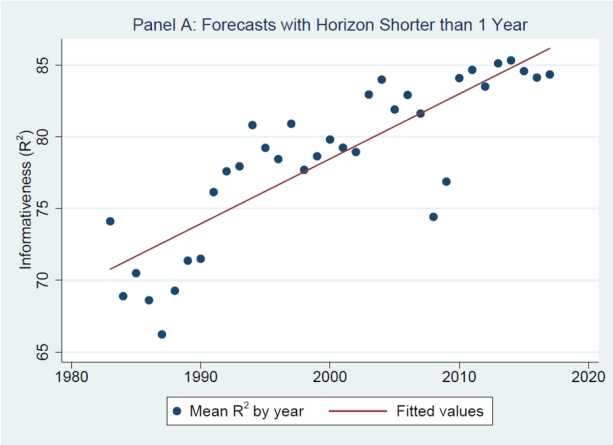
© 2025 – 2026 Chuan Shi | Alternative Data and Applications

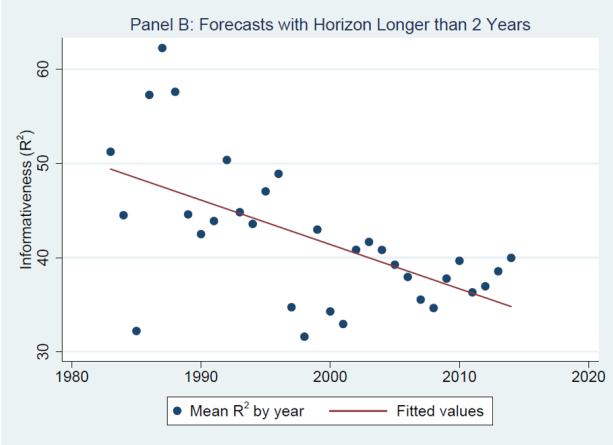
- The slope of the red curve is steeper than that of the blue curve.
- After 2000 (when alternative data began to gradually emerge in the U.S. stock market), **forecast accuracy improved over short horizons, but at the cost of reduced accuracy over longer horizons.**
- The mixed impact is still unknown!



出处：Dessaint, Foucault, and Fresard (2022)

© 2025 – 2026 Chuan Shi | Alternative Data and Applications





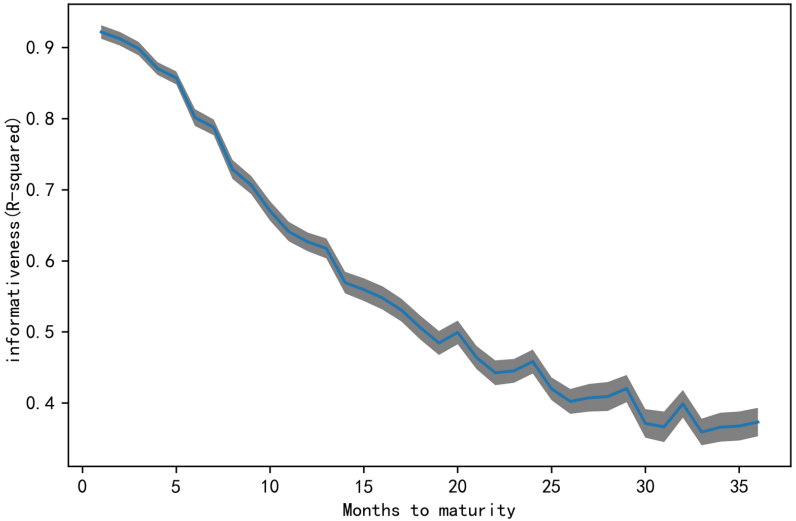
出处： Dessaint, Foucault, and Fresard (2022)

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Term Structure of Forecast Accuracy – China Evidence

5 Implication for Financial Forecast

- Data source: Analyst forecast data from Suntime (Zhao Yang Xiong Xu).
- Data spans from 2012 to 2023, with a total of 4,208,520 forecast samples.

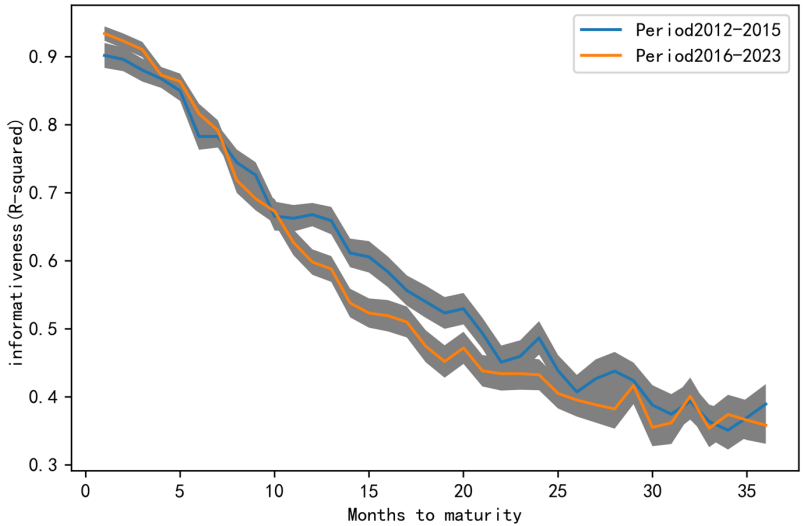


© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Change of Term Structure – China Evidence

5 Implication for Financial Forecast

- Divided the analyst earnings forecast samples into two periods, before and after 2016.
- The accuracy of short-term forecasts indeed improved in the later period.
- In the 2-year forecast horizon, the accuracy in the second period significantly decreased.
- However, there is no significant difference between the two periods for forecast horizons beyond 2 years.



© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- **Do NOT equate alternative data with anomaly/factor research!**
- More important questions to be addressed with the help of alternative data:
  - How can alternative data help us understand the systematic risks driving asset returns across time and cross-sections?
  - How does alternative data influence the behavior of various financial participants?
  - What is the dynamic relationship between data richness and the informational content in asset prices over time?
  - How does alternative data impact investor learning, and could it lead to false predictability?
- Alternative data in asset pricing goes beyond just uncovering anomalies.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Table of Contents

6 Use Case: Patent Data in China

- Introduction
- Alternative Data and Quantitative Investment
- (Pseudo-)Alternative Data – Economic Links
- (Pseudo-)Alternative Data – Fund-Implied Information
- Implication for Financial Forecast
- Use Case: Patent Data in China

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Research Goal: Uncover predictive information about asset pricing contained in patent data for the Chinese stock market.
- Data source: **QuantData** (https://www.quantdata.com.cn), a leading alternative data vendor in China. They provide high-quality, long-span patent data.
- To measure similarity between firms, we need to find their patent distributions over different patent topics.
- Motivated by Bybee et al. (forthcoming) on news, we first tried to classify patents into different topics by using textual analysis, e.g., Latent Dirichlet allocation (LDA).

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Challenges with LDA for Patent Texts

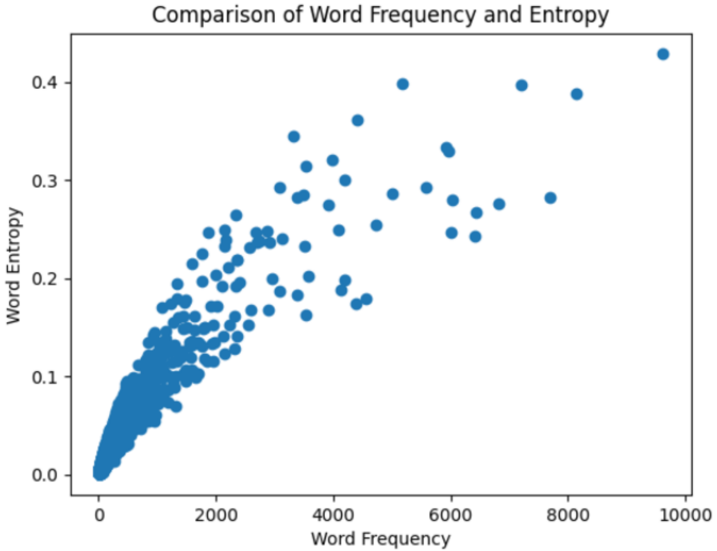
- Preprocessing Steps:
  - Tokenize abstract data from patents.
  - Remove stop words.
  - Create a dictionary and transform text into a document-word matrix.
- Key Challenge:
  - **Domain-specific terms** (e.g., “device”, “system”) that do not contribute to classification.

```
(0,
 '0.039*"装置" + 0.009*"信号" + 0.009*"结构" + 0.008*"系统" + 0.008*"表面" + 0.007*"设置" '
 '+ 0.005*"设备" + 0.005*"部分" + 0.005*"特征" + 0.004*"步骤"'),
(1,
 '0.015*"装置" + 0.015*"检测" + 0.011*"信号" + 0.011*"设置" + 0.006*"工艺" + 0.005*"技术" '
 '+ 0.005*"结构" + 0.005*"金属" + 0.005*"信息" + 0.005*"催化剂"'),
(2,
 '0.012*"装置" + 0.010*"系统" + 0.010*"数据" + 0.008*"方向" + 0.006*"结构" + 0.005*"定位" '
 '+ 0.005*"电路" + 0.005*"设置" + 0.005*"基板" + 0.005*"单元"'),
(3,
 '0.010*"装置" + 0.007*"部件" + 0.007*"材料" + 0.006*"区域" + 0.006*"系统" + 0.005*"部分" '
 '+ 0.005*"步骤" + 0.005*"驱动" + 0.005*"结构" + 0.005*"设置"'),
(4,
 '0.037*"装置" + 0.014*"数据" + 0.009*"图像" + 0.009*"设置" + 0.009*"信息" + 0.008*"移动" '
 '+ 0.007*"组件" + 0.007*"单元" + 0.007*"系统" + 0.007*"信号"'),
(5,
 '0.013*"装置" + 0.012*"设置" + 0.011*"部分" + 0.010*"表面" + 0.009*"驱动" + 0.009*"系统" '
 '+ 0.007*"元件" + 0.006*"电极" + 0.006*"材料" + 0.006*"电路"'),
(6,
 '0.016*"装置" + 0.011*"模块" + 0.009*"信息" + 0.009*"单元" + 0.008*"系统" + '
 '0.008*"传感器" + 0.007*"设置" + 0.007*"检测" + 0.007*"测试" + 0.006*"电极"'),
```


© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Remove Domain-Specific Terms

- Identify and remove non-informative terms using frequency analysis or information entropy.
- In patent texts, since no word dominates with very high frequency, entropy and word frequency exhibit a roughly proportional relationship.
- Examples of filtered terms: “system”, “process”, “structure”, “data”, etc.



© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Improving LDA Results

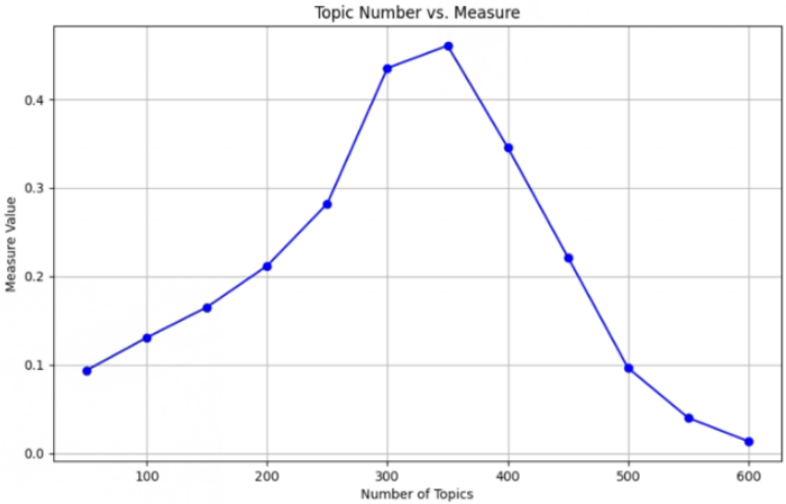
6 Use Case: Patent Data in China

- Post-filtering LDA results produced more meaningful keywords for topics.
- Example: Keywords after filtering align better with distinct themes.
- Empirical challenge: need to determine number of domain-specific terms to be removed.

```
[(0,
  '0.004*"总线" + 0.004*"排气" + 0.003*"工作台" + 0.002*"电容器" + 0.002*"光栅" + '
  '0.002*"石墨" + 0.002*"晶片" + 0.002*"主轴" + 0.002*"抗体" + 0.002*"波束"'),
 (1,
  '0.003*"绕组" + 0.003*"文本" + 0.003*"图象" + 0.003*"刀具" + 0.002*"指纹" + 0.002*"滑槽" '
  '+ 0.002*"探针" + 0.002*"解码" + 0.002*"链路" + 0.002*"软管"'),
 (2,
  '0.004*"标签" + 0.003*"滚筒" + 0.003*"协议" + 0.003*"矩阵" + 0.003*"调度" + 0.002*"织物" '
  '+ 0.002*"路由" + 0.002*"石墨" + 0.002*"纤维素" + 0.002*"环氧树脂"'),
 (3,
  '0.004*"任务" + 0.003*"引线" + 0.003*"晶片" + 0.003*"事件" + 0.003*"光盘" + 0.002*"字符" '
  '+ 0.002*"错误" + 0.002*"凸轮" + 0.002*"曝光" + 0.002*"基片"'),
 (4,
  '0.004*"样本" + 0.004*"液晶" + 0.003*"人脸" + 0.003*"媒体" + 0.003*"检测器" + '
  '0.003*"离合器" + 0.002*"网关" + 0.002*"导流" + 0.002*"评估" + 0.002*"应用程序"'),
 (5,
  '0.007*"时钟" + 0.005*"盖板" + 0.003*"沟槽" + 0.003*"液晶" + 0.003*"滤波器" + '
  '0.002*"校准" + 0.002*"道路" + 0.002*"镜头" + 0.002*"连杆" + 0.002*"继电器"'),
 (6,
  '0.004*"燃料电池" + 0.003*"影像" + 0.003*"过滤器" + 0.003*"电缆" + 0.002*"转矩" + '
  '0.002*"氢气" + 0.002*"连接件" + 0.002*"等离子体" + 0.002*"螺旋" + 0.002*"活塞杆"'),
```


© 2025 – 2026 Chuan Shi | Alternative Data and Applications

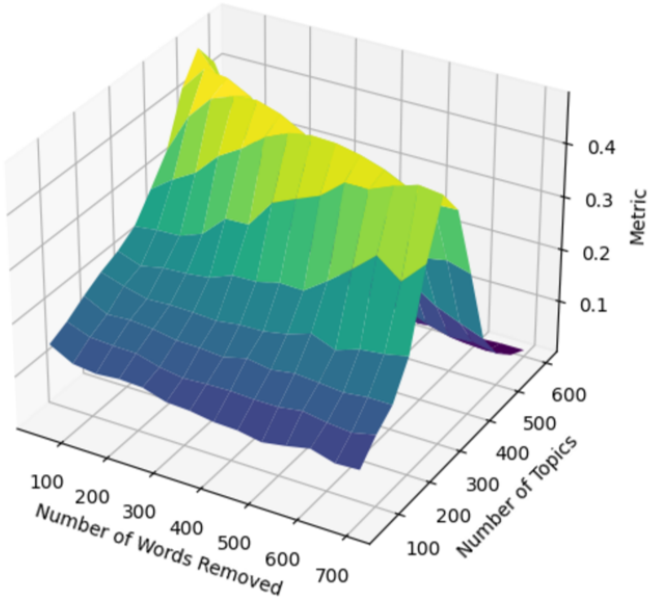
- Use Jensen–Shannon (JS) divergence to determine the optimal number of topics.
- JS divergence measures the overall distinction between topics.
- Optimal number of topics found around 350 (with 500 domain-specific terms removed).



© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Robustness Check

- Analyze the relationship between the number of removed terms and the optimal number of topics.
- As more terms are removed, optimal topics decrease from 500 to around 300.



© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Validation of LDA Using IPC Subclass

- Use IPC subclasses as prior to verify the reliability of LDA.
- Both LDA and IPC categorize each patent. We calculate patent similarity based on these classifications.
- For each patent, identify the top N most similar patents from both methods, then compute the Jaccard similarity between these sets.
- For most patents, the Jaccard similarity is **zero**, indicating a significant difference in similarity calculation between LDA and IPC classifications.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Validation of LDA Using IPC Subclass

- Use IPC subclasses as prior to verify the reliability of LDA.
- Both LDA and IPC categorize each patent. We calculate patent similarity based on these classifications.
- For each patent, identify the top N most similar patents from both methods, then compute the Jaccard similarity between these sets.
- For most patents, the Jaccard similarity is **zero**, indicating a significant difference in similarity calculation between LDA and IPC classifications.

LDA classification lacks alignment with the IPC subclass benchmark.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Large Language Model for Patent Classification

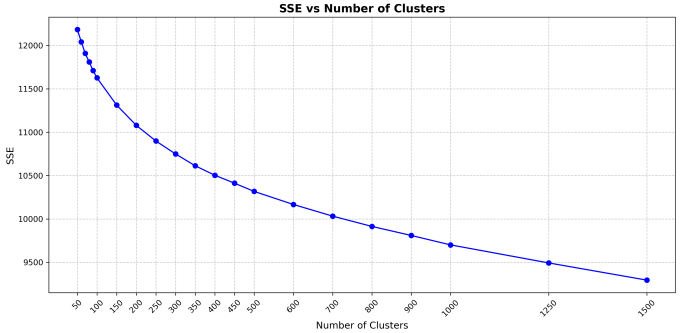
6 Use Case: Patent Data in China

- Use large language models (LLM) to construct semantic vectors for patent abstract, improving classification accuracy.
- Traditional models like LDA lack semantic understanding for accurate classification.
- Chose **BGE (BAAI General Embedding)** bge-large-zh-v1.5 model due to its strengths in capturing semantic context in text. **Unlike word vector models, BGE considers the context of words in the text.**
- BGE outperforms similar models (e.g., OpenAI's text embedding 002) in semantic retrieval for both Chinese and English.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

Methodology for Patent Classification

- Process each patent abstract using BGE to create 1024-dimensional semantic vectors.
- Text exceeding 520 characters is truncated to fit model requirements, though most abstracts are within this limit.
- Apply K-means clustering on semantic vectors to classify patents.
- The optimal number of clusters ($N = 500$) was determined using the elbow method.



© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Practical Considerations and Validation with IPC

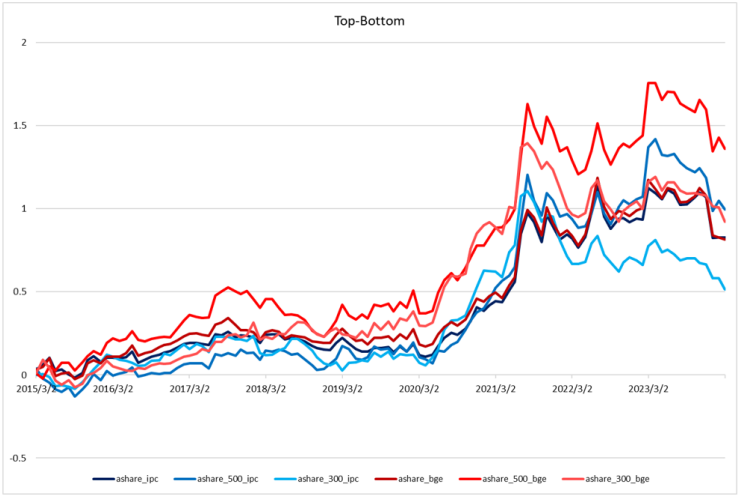
- Due to a large dataset (over 6 million patents), clustering was performed on a 100,000-patent sample.
- Remaining patents and new entries are mapped to pre-defined categories.
- Used IPC subclasses as a benchmark to assess classification quality.
- Calculated Jaccard similarity between BGE and IPC classifications. BGE classifications align more closely with IPC.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

T-Link Momentum in China

6 Use Case: Patent Data in China

- Aggregated results at the company level, creating a 500-dimensional topic vector per company.
- Calculate similarity between firms' patent vectors to construct the T-link variable.
- Portfolio sort results show the effectiveness of the variable.



© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Fama-MacBeth Regression

6 Use Case: Patent Data in China

- Test effectiveness of T-Link while controlling for common Barra risk factors.
- The following table reports Fama-MacBeth regression t-statistics.

	BGE	IPC	Both
Panel A: All			
BGE	3.07		2.70
IPC		3.20	3.05
Panel B: CSI500			
BGE	3.53		2.64
IPC		3.38	2.05

© 2025 – 2026 Chuan Shi | Alternative Data and Applications

- Test effectiveness of T-Link while controlling for common Barra risk factors.
- The following table reports Fama-MacBeth regression t-statistics.

	BGE	IPC	Both
Panel A: All			
BGE	3.07		2.70
IPC		3.20	3.05
Panel B: CSI500			
BGE	3.53		2.64
IPC		3.38	2.05

Both factors add value in predicting expected returns. More importantly, they capture different dimensions of patent data.

© 2025 – 2026 Chuan Shi | Alternative Data and Applications



Alternative Data and Applications

Questions?

© 2025 – 2026 Chuan Shi | Alternative Data and Applications